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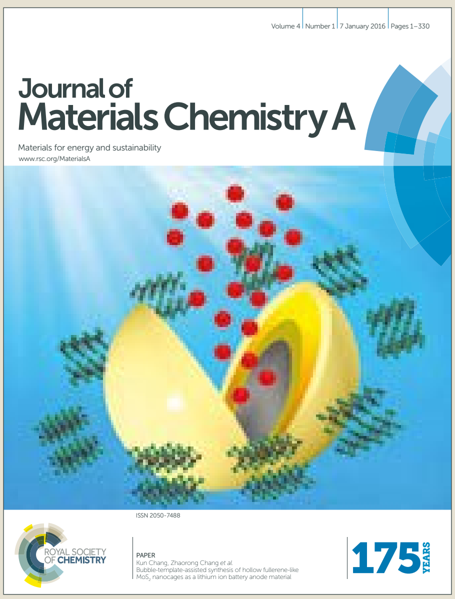

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ARTICLE

Highly efficient and ultra-stable boron-doped graphite felt electrodes for vanadium redox flow batteries

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H.R. Jiang, W. Shyy, L. Zeng, R.H. Zhang, T.S. Zhao*

Developing high-performance electrodes with high operating current density and long-term cycling stability is crucial to the widespread application of vanadium redox flow batteries (VRFBs). In this work, boron-doped graphite felt electrodes are designed, fabricated and tested for VRFBs. First-principles study firstly demonstrates that the boron-doped carbon surface possesses highly active and stabilized reaction sites. Based on this finding, we fabricate boron-doped graphite felt electrodes for VRFBs. Testing results show that the batteries with boron-doped graphite felt electrodes achieve energy efficiencies of 87.40% and 82.52% at the current densities of 160 and 240 mA cm^{-2}, which are 15.63% and 19.50% higher than those with the original electrodes. In addition, the batteries can also be operated at high current densities of 320 and 400 mA cm^{-2} with energy efficiencies of 77.97% and 73.63%, among the highest performances in the open literature. More excitingly, the VRFBs with the boron-doped graphite felt electrodes exhibit excellent stability during long-term cycling tests. The batteries can be stably cycled for more than 2000 cycles at 240 mA cm^{-2} with ultra-low capacity and efficiency decay rates of only 0.028% and 0.0002% per cycle. In addition, after refreshing the electrolytes, the performances of the batteries are nearly recovered regardless of the evitable decay of the membrane. All these results suggest that the highly efficient and ultra-stable boron-doped graphite felts are promising electrodes for VRFBs.

1. Introduction

Developing efficient, cost-effective and reliable large-scale energy storage systems is continuously attracting attentions from both academic and industrial fields, primarily due to the intermittent and fluctuated nature of renewable energies (*e.g.,* solar power, wind power), which would destabilize the energy grid if directly used [1-8]. However, the well-known lithium-ion batteries that were firstly commercialized by Sony in 1991, fail to satisfy all the requirements because of their poor scalability, limited design flexibility and safety concerns [9-17]. Fortunately, the redox flow battery, which possesses the advantages including decoupled energy and power, high energy efficiency, excellent reliability, fast response and long cycle life, offers a more promising choice for large-scale energy storage [18-24]. Among the state-of-art redox flow batteries, the vanadium redox flow batteries (VRFBs) arouse the most interests for commercial application, attributed to the same element vanadium is adopted in both negative and positive electrolytes. Therefore, the severe cross-contamination issue for flow batteries is alleviated in VRFBs [25]. Even though, the current VRFBs still suffer from the issues such as low energy efficiencies and short lifetime, leading to the high capital cost for the whole systems [26].

As a key component for VRFBs, the electrode plays a dominant role in determining the battery performance, because it not only provides the active sites for redox reactions, but also affects the ion/mass transport inside its porous structure. Typically, the graphite felt is used as the electrode materials for VRFBs due to its high electronic conductivity, good corrosion resistance, large porosity and low cost. However, the original graphite felt electrode shows poor kinetics towards vanadium redox reactions, resulting in the poor performance in the battery tests. In this regard, tremendous efforts have been taken to modify the original graphite felt to enhance its performance. Generally, the methods can be divided into two types: one is depositing catalysts on the surfaces of carbon fibers, such as Bi [27], B_4C [29], ZrO_2 [30], TiN [31], B/N-co-doped porous carbon [32] and graphene oxide [33]; the other is etching the surfaces by chemical or electrochemical methods, such as thermal treatment [34], KOH activation [35], water activation [36] and electrochemical activation [37]. With these efforts, the hydrophily, permeability and activity of the graphite felts are greatly increased, and the assembled VRFBs exhibit a good rate capability that can reach an energy efficiency of higher than 80% at the current density of 160 mA cm^{-2} [29].

Unfortunately, compared with the great success in enhancing the rate capability of VRFBs, less attention has been paid to improve battery's cycling stability. Considering the fact that VRFBs are developed for long-term and large-scale energy storage, the cycling stability plays an equal or more important role in determining the performances of VRFBs [38]. The fast decay in energy efficiency and capacity would not only waste the precious active materials, but also increase the burden to maintenances, hindering the long-term operation and increasing the capital cost. Although previous efforts can effectively enhance the rate capability of VRFBs, most of them fail to satisfy the requirements for long-term operation with stable

Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong, China.
**E-mail: metzhao@ust.hk*

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energy efficiency and high capacity retention rate. For example, the deposited catalysts are easily aggregated and dislodged by the flowing electrolyte, inevitably leading to the loss of active surface area [29, 32]. Meanwhile, the oxygen-functional groups, which are the typical active sites for etched graphite felt, are not stable during repeated charge and discharge processes, deteriorating the batteries' cycling performances [39-41]. In this regard, some researchers try to enhance the cycling stability by increasing the binding of nanoparticles on carbon fibers. For example, Ji *et al.* [42] adopted carbonized polydopamine (PDA) as the support on electrodes to bind Mn_3O_4 nanoparticles, and results showed increased cycling stability than the original electrode without PDA. However, the dislodgement of nanoparticles under flowing electrolyte cannot be totally avoided, and the long-term cycling tests are still missing. Another strategy to enhance the stability of electrodes is replacing unstable oxygen-functional groups with other more stable active sites by heteroatom-doping, such as nitrogen-doping [43] and phosphate-doping [44]. Recently, Huang *et al.* [45] prepared phosphorus and fluorine co-doped graphite felt electrode for VRFBs, and their assembled batteries can be operated for 1000 cycles at the current density of 120 mA cm^{-2} with an energy efficiency of 79.2%. Even progress had been made, the battery also suffered from a ~3% energy efficiency decay and ~50% capacity decay after 1000 cycles. Meanwhile, the in-depth understanding on how the heteroatom-doping enhances the stability of graphite felt electrode is still not clear. More importantly, the rate capabilities of the state-of-art heteroatom-doped graphite felts are incomparable with other high-performance electrodes, greatly limiting their further widespread applications.

To address these issues, in this work, highly efficient and ultra-stable boron-doped graphite felts are rationally designed to act as the electrodes for VRFBs. First-principles study demonstrates that the boron-doped carbon surface possesses highly active and stabilized reaction sites. In the battery tests, the assembled VRFB with boron-doped graphite felt electrodes achieves an energy efficiency of 82.51% at a high current density of 240 mA cm^{-2}, which is 19.5% higher than that with original electrode. More remarkably, the battery can be stably cycled for more than 2000 cycles with a high capacity retention rate of 99.972% per cycle and high energy efficiency retention rate of 99.9998% per cycle, demonstrating the excellent stability during long-term cycling.

2. Methods

2.1 Computational methods

Density functional theory (DFT) based first-principles studies were conducted by adopting ABINIT [46] code. The generalized gradient approximation (GGA) of Perdew-Burke-Ernzerhof (PBE) type [47] was adopted to depict the exchange correlation interaction. The projector augmented wave (PAW) method [48] was used to deal with core electrons. The convergence tests were firstly carried out and an energy cutoff of 22 Ha was chosen for the plane wave-basis expansion. To avoid the error caused by the periodicity, vacuum layers of 20 Å were used for all calculations. The slab models were built according to a monolayer graphite (0001) surface in 4×4×1 supercell with 4×4×1 k-point Monkhorst-Pack grids [49], the feasibility of which had been proved by previous work [50]. The force tolerance of self-consistent-field cycles was 4.0×10^{-5} Ha $Bohr^{-1}$ and that of the structural optimization was 4×10^{-4} Ha $Bohr^{-1}$.

The energy barriers to remove oxygen-functional group and boron atom were adopted to evaluate the stability of the corresponding carbon surfaces, which can be calculated by:

$$E_{barrier} = E_{system} - E_{group} - E_{substrate} \tag{1}$$

where E_{system} is the DFT energy of boron-doped or oxygen-functionalized carbon surface; E_{group} is the DFT energy of an oxygen-functional group or boron atom; $E_{substrate}$ is the DFT energy of the carbon surface after removing hydroxyl group or boron atom.

2.2 Experimental methods

The commercial graphite felt (SGL carbon, GFA6 EA) was separated along cross-section into two pieces, washed with ethanol, dried in a vacuum oven and adopted as the original electrode material. To increase the wettability, the graphite felts (GFs) were pretreated at 500 °C for 5 hours in the ambient air with a heating rate of 5 °C min^{-1}. Then, the thermally treated graphite felts (thermally treated GFs) were soaked in H_3BO_3 (Sigma-Aldrich, 99.8%) solutions with varying concentrations of 1 M, 2M and 3 M. After that, the formed mixtures were dried in the vacuum oven and heated in a tube furnace under nitrogen gas flow at 800 °C for 90 minutes to fabricate the boron-doped graphite felts (B-doped GFs). After washing in deionized water and drying in vacuum oven, the final samples were obtained and named as B-doped GF-1, B-doped GF-2 and B-doped GF-3 according to the concentration of H_3BO_3 solutions.

Cyclic Voltammetry (CV) tests were carried out on the Autolab (PGSTAT30) workstation adopting a typical three-electrode cell in a solution containing 0.1 M VO^{2+} + 3 M H_2SO_4 at a scan rate of 10 mV s^{-1}. The GF, thermally treated GF and B-doped GF-2 were fixed to the glassy carbon by a homemade Polytetrafluoroethylene (PTFE) tubing and acted as the working electrodes, as shown in Fig. S1. Saturated calomel electrode (SCE) and platinum mesh were adopted as reference and counter electrodes, respectively. The voltage windows of -0.7 to -0.2 V and 0.6 to 1.1 V *vs.* SCE were used for V^{2+}/V^{3+} and VO^{2+}/VO_2^{+} redox reactions. The electrochemical double-layer capacitances were tested by immerging the electrodes into solution containing 3 M H_2SO_4 at a scan rate of 200 mV s^{-1} with the voltage window of 0.4-0.9 V *vs.* SCE.

The battery tests were all carried out on the Arbin BT2000 (Arbin Instrument, Inc.) with a homemade single cell with serpentine flow field. The cutoff voltages for charge and discharge were 1.65 and 0.8 V, respectively. Graphite felts with the active area of 2.0 cm × 2.0 cm were adopted as both negative and positive electrodes, and were separated by a commercial Nafion® 212 (Dupont, USA) membrane. The compression ratio of electrodes was 60%. The negative electrolyte was 20 mL solution containing 1 M V^{3+} + 3 M H_2SO_4 and the positive electrolyte was 20 mL solution containing 1 M VO^{2+} + 3 M H_2SO_4. Both electrolytes were circulated at a fixed flow rate of 46 mL min^{-1} by a 2-channel peristaltic pump (Longer pump, WT600-2J). Nitrogen gas was bubbled to expel the air from electrolytes and reservoirs to avoid the undesirable side reactions. The X-ray photoelectron spectroscopy (XPS) characterizations were carried out by a Physical Electronics PHI 5600 multi-technique system using Al monochromatic X-ray at a power of 350 W. The

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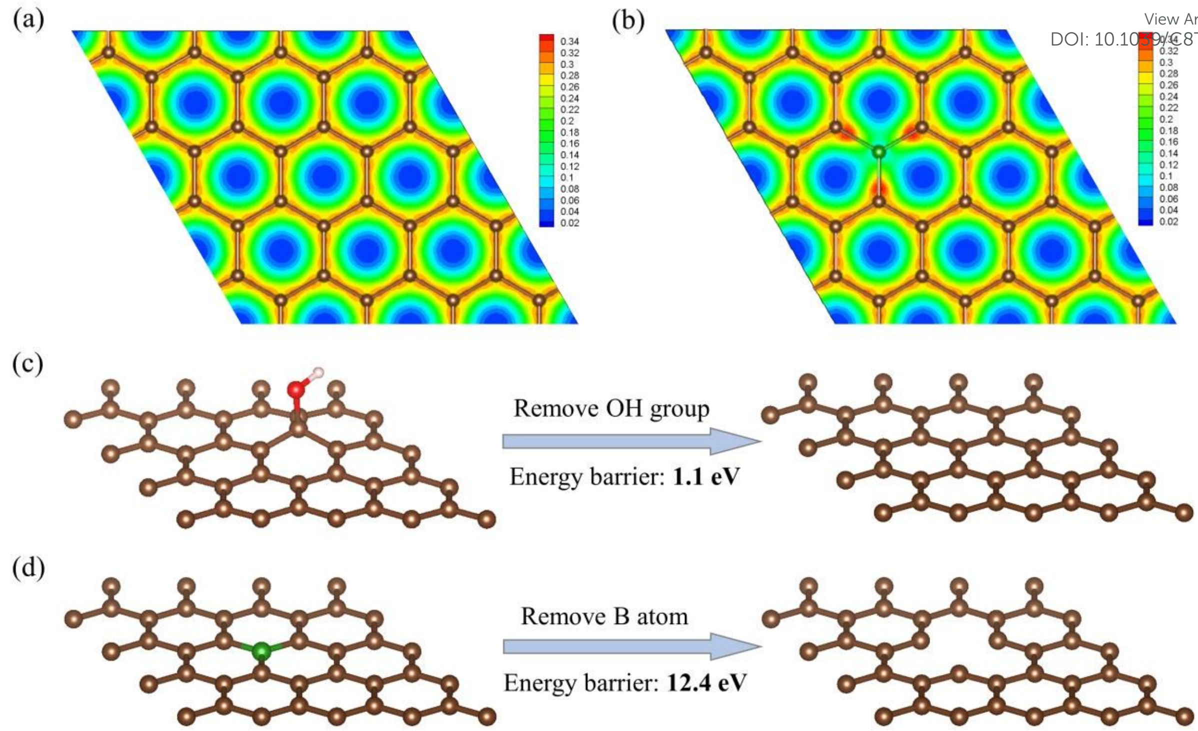

Fig. 1. The optimized structures and the corresponding electron density contours of (a) pristine graphite (0001) surface and (b) boron-doped graphite (0001) surface. The energy barriers to remove (c) OH group and (d) boron atom from carbon surfaces. The brown, green, red and white balls represent for carbon, boron, oxygen and hydrogen atoms, respectively.

surface morphology was observed by a scanning electron microscope (JEOL-6390 SEM).

3. Results and discussion

To have an in-depth understanding on the high activity and stability of boron-doped graphite felt for vanadium redox reactions, the first-principles studies are firstly carried out to clarify the underneath mechanism. The optimized structures and the corresponding electron density contours of pristine and boron-doped graphite (0001) surfaces are shown in Figs.1a and b. Because the detailed mechanisms of V^{2+}/V^{3+} and VO^{2+}/VO_2^{+} redox reactions in vanadium redox flow batteries are still unknown [9, 51], here, we try to exhibit the high efficiency and high stability of the boron-doped graphite felt based on its intrinsic properties from some well-developed theories [52-54]. It is seen that both pristine and boron-doped graphite (0001) surfaces are planar with all the atoms locating at the same atomic layer, indicating the existence of boron atom would not deform the original carbon surface, which is different from other heteroatom-doping such as phosphorus-doping [55] and sulfur-doping [56]. The optimized C-C bond length in pristine graphite (0001) surface is 1.43 Å, shorter than that of C-B bond in boron-doped graphite (0001) surface of 1.49 Å, attributed to the larger radius of boron atom than carbon atom. Results also show that in pristine graphite (0001) surface, the electrons are localized between C-C bonds and the electrons' distribution are identical for all the bonds, which would passivate the activity of electrons towards redox reactions [57]. On the contrary, in the electron density contour of boron-doped graphite (0001) surface, it is clearly seen that a high electron density area is formed around the carbon atoms bonded to boron atom and a low electron density area is formed near the boron atom, attributed to the fact that electronegativity of carbon atom is higher than boron atom. Therefore, the existence of boron atom in pristine graphite (0001) surface enables to effectively activate the inert electrons, which would benefit the redox reactions and enhance the activity of the surface. Based on the above results, it is predicted that the boron-doped graphite felt can achieve higher activity than the original graphite felt, and thus leads to a higher performance when adopted as the electrodes for VRFBs.

To evaluate the stability of boron-doped carbon surface compared with the traditional oxygen-functionalized carbon surface, the energy barriers to remove an oxygen-functional group or boron atom from carbon surfaces are calculated and compared, as shown in Figs. 1c and d. Here, the hydroxy group is chosen to be the representative oxygen-functional group, because it is demonstrated to be the most active reaction site for vanadium redox reactions in oxygen-functionalized graphite felt [58]. It is interesting to find that the energy barrier to remove the hydroxy group from carbon surface is as low as 1.1 eV, indicating the oxygen-functionalized graphite (0001) surface is thermodynamically not stable enough. These results also mean that, when adopting the oxygen-functionalized graphite felts as the electrodes in VRFBs, the oxygen-functional groups are easy to be removed during repeated charge and discharge processes, deteriorating the stability of electrodes and reducing the cycling performances of batteries, which is consistent with previous experimental results [40, 41]. On the contrary, it is also found that the energy barrier to remove the boron atom from the carbon surface is 12.4 eV, which is an order of magnitude larger than that of removing the hydroxy group, suggesting the boron-doped carbon surface is

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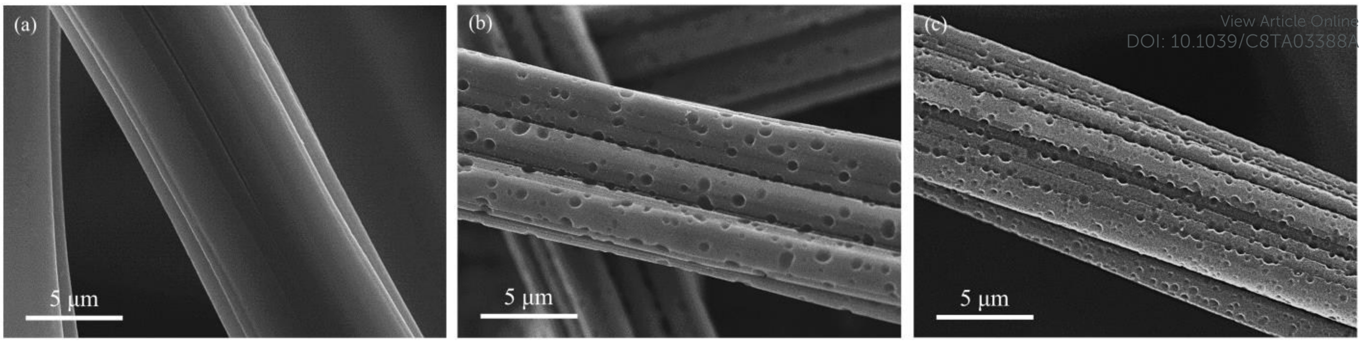

Fig. 2. SEM images of (a) original graphite felt, (b) thermally treated graphite felt and (c) boron-doped graphite felt.

much more stable than the oxygen-functionalized carbon surface. Another important finding is that, unlike the oxygen-functionalized carbon surface which forms pristine graphite (0001) surface after removing oxygen-functional groups, the single vacancy is formed after removing the boron atom from boron-doped carbon surface, as shown in Fig. 1d. Similar to the boron-doped carbon surface, the single vacancy can also lead to the non-uniform distribution of electrons because of the existence of dangling carbon atoms [59]. Therefore, even if the boron atom is lost, the high activity of the carbon surface can still be largely maintained. In this regard, the boron-doped graphite felt is also expected to have higher stability than the traditional oxygen-functionalized graphite felt, corresponding to the much better cycling performances for VRFBs.

Based on these computational results and analysis, the boron-doped graphite felts are thus rationally designed and fabricated. SEM images of original graphite felt, thermally treated graphite felt and boron-doped graphite felt (B-doped GF-2) are shown in Fig. 2. It is seen that the surface of carbon fibers in original graphite felt is very smooth with seldom defects on it, but that in boron-doped graphite felt surface is much rougher with many pores existing, which increases the effective surface areas and benefits the battery performance. The calculated electrochemical double-layer capacitances for GF, thermally treated GF and boron-doped GF are 8, 589 and 660 mF cm^{-2}, respectively, as shown in Fig. S2. Therefore, the enhanced surface area on boron-doped graphite felt are attributed to two factors: the thermal pretreatment and the doping process.

To evaluate the electrochemical performances of boron-doped graphite felt, CV tests are carried out. The comparison of original, thermally treated and boron-doped graphite felts towards V^{2+}/V^{3+} and VO^{2+}/VO_2^+ redox reactions are presented in Figs. 3a and b, respectively. It is found that, among the samples studied, the boron-

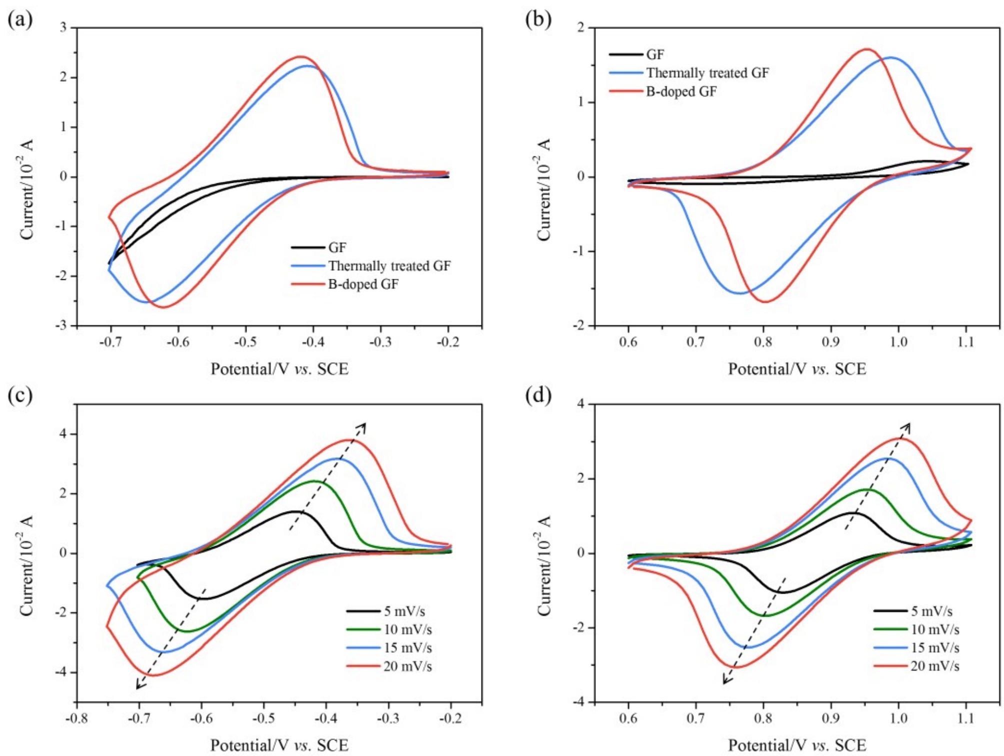

Fig. 3. CV curves of original, thermally treated and boron-doped graphite felts with the potential windows of (a) -0.7 to -0.2 V and (b) 0.6 to 1.1 V at the scan rate of 10 mV s^{-1}. CV curves of boron-doped graphite felt for (c) V^{2+}/V^{3+} and (d) VO^{2+}/VO_2^+ redox reactions at various scan rates.

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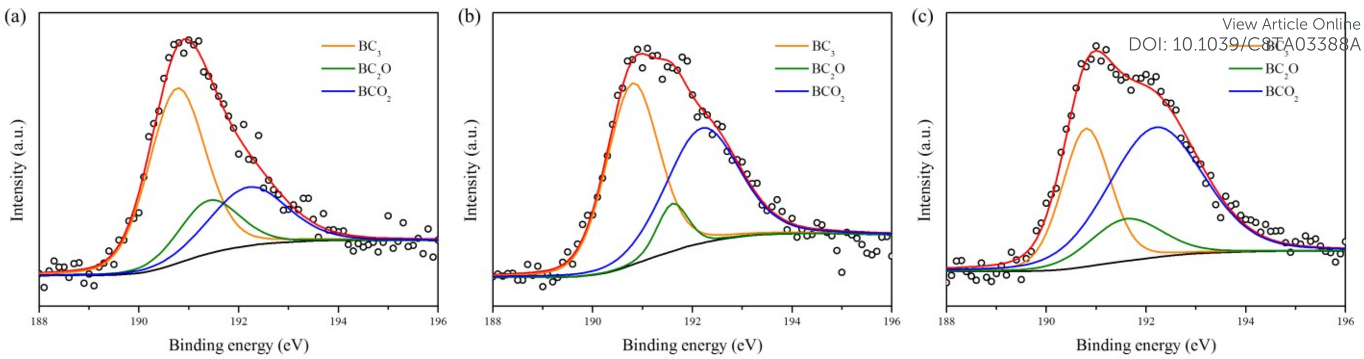

Fig. 4. B1s core-level spectra of (a) B-doped GF-1, (b) B-doped GF-2 and (c) B-doped GF-3.

doped graphite felt exhibits the highest peak current and the lowest peak potential separations, indicating the best electrochemical performances. On the original graphite felt, there is no obvious peaks observed for V^{2+}/V^{3+} and VO^{2+}/VO_2^+ redox reactions, because of its poor activity and hydrophobic property. The electrochemical performances of graphite felt can be enhanced after thermal treatment, and the thermally treated graphite felt shows a negative peak potential separation of 239 mV and a positive peak potential separation of 220 mV. More importantly, after doping boron atoms, the peak potential separations for negative and positive sides continuously decrease to 205 and 151 mV, demonstrating the high electrochemical performance of boron-doped graphite felt. The CV curves of boron-doped graphite felt at various scan rate for V^{2+}/V^{3+} and VO^{2+}/VO_2^+ redox reactions are shown in Figs. 3c and d, respectively, and the obvious oxidation and reduction peaks are observed at all scan rates. In addition, the peak currents and the square root of the scan rate are found to have linear relationships, indicating the reactions are controlled by transport in the scan rate range.

In order to further optimize the fabrication process, the boron-

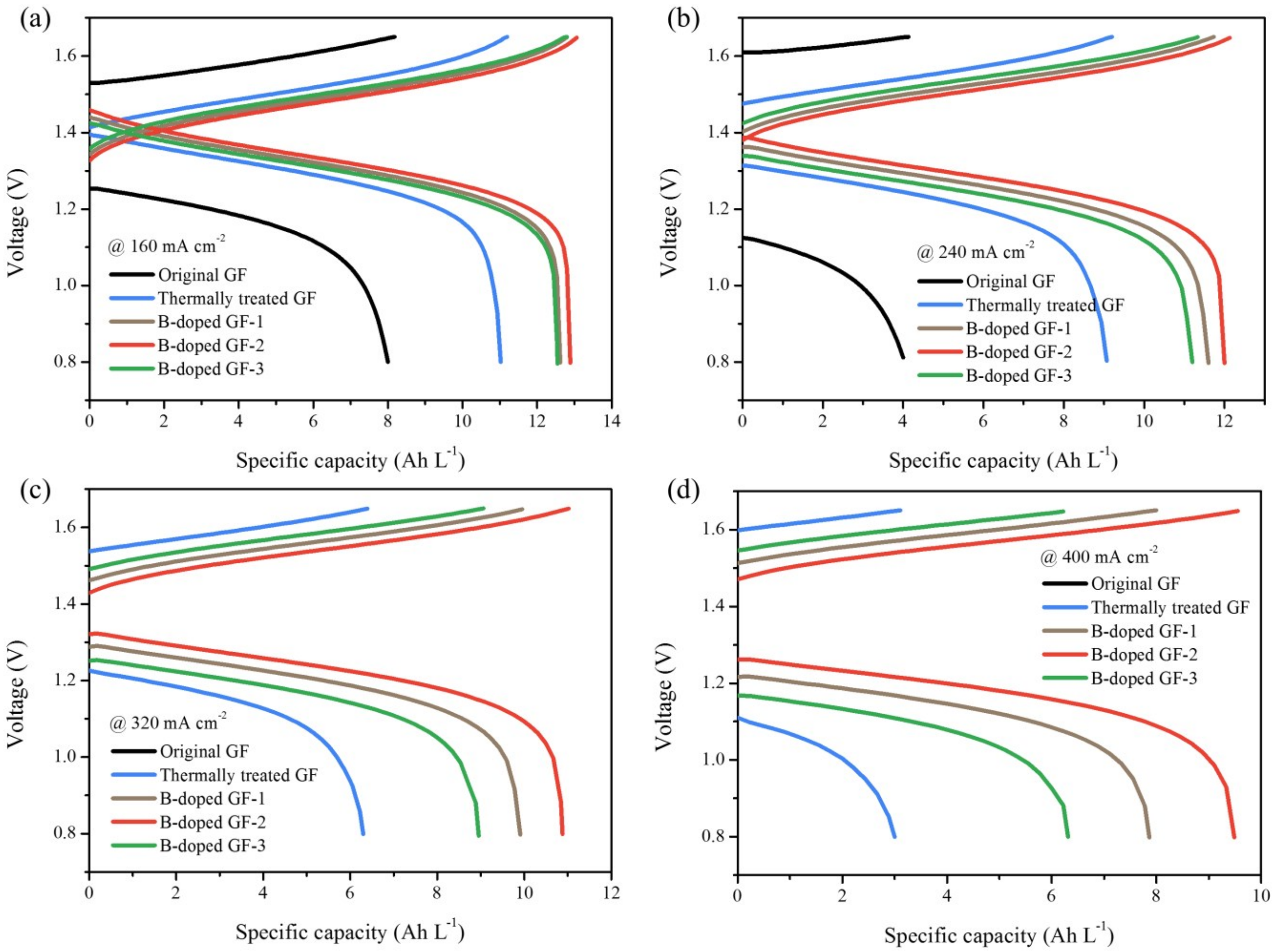

Fig. 5. The charge-discharge curves at the current densities of (a) 160, (b) 240, (c) 320 and (d) 400 mA cm^{-2} for VRFBs with different electrodes.

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Table 1. Elemental composition and distribution of types of boron bonds on thermally treated GF (TGF), B-doped GF-1, B-doped GF-2 and B-doped GF-3.

Samples	C 1s (%)	B 1s (%)	O 1s (%)	BC_3 (%)	BC_2O (%)	BCO_2 (%)
TGF	92.43	-	7.57	-	-	-
B-doped GF-1	94.5	0.61	4.89	0.34	0.11	0.16
B-doped GF-2	93.72	0.82	5.46	0.40	0.07	0.35
B-doped GF-3	93.16	0.84	6.00	0.25	0.12	0.47

doped graphite felts soaked with various H_3BO_3 concentrations are prepared and tested. The XPS characterizations are conducted to analyze the surface properties of various samples. The B1s core-level spectra of B-doped GF-1, B-doped GF-2 and B-doped GF-3 are shown in Fig. 4a-c, and the atomic fractions of C, B and O are summarized in Table 1. It is found that atomic fractions of boron for B-doped GF-1, B-doped GF-2 and B-doped GF-3 are 0.61%, 0.82% and 0.84%, respectively. Here, the B 1s core-level spectra can be deconvoluted into three peaks at binding energies of 190.8, 191.6 and 192.3 eV, corresponding to BC_3, BC_2O and BCO_2 bonds [60, 61]. It is seen that, as the concentrations of H_3BO_3 solutions increase, the contents of oxygen-functional groups and BCO_2 bonds increase. In addition, the contents of BC_3 bonds for B-doped GF-1, B-doped GF-2 and B-doped GF-3 are 0.34%, 0.40% and 0.25%, respectively.

Then, the real performances of boron-doped graphite felt electrodes are tested in a single cell. The charge-discharge curves for VRFBs with original graphite felt, thermally treated graphite felt, B-doped GF-1, B-doped GF-2 and B-doped GF-3 at the current densities of 160, 240, 320 and 400 mA cm^{-2} are shown in Figs. 5a-d, respectively. It is found that the VRFBs with boron-doped graphite felt electrodes have lower charge/discharge overpotentials and higher capacities than those with the original and thermally treated graphite felt electrodes, demonstrating the enhanced performances of graphite felts through boron-doping, which is consistent with the CV results. In addition, the batteries can also be operated at the high current densities of 320 and 400 mA cm^{-2}. Another interesting finding is that, for the boron-doped graphite felt electrodes, the overpotentials decrease in the sequence of B-doped GF-3 > B-doped GF-1 > B-doped GF-2, which means the B-doped GF-2 has the highest performances among samples studied. Recall that in the XPS results, B-doped GF-2 possesses the highest content of BC_3 bond and B-doped GF-3 has the highest content of oxygen-functional groups among the prepared boron-doped graphite felts. Therefore, it is expected that BC_3 bond instead of the oxygen-functional group is the main reactive sites for boron-doped graphite felt and the contents of BC_3 bonds play the key role in determining the performances of VRFBs.

The coulombic efficiency, voltage efficiency and energy efficiency for VRFBs with original graphite felt, thermally treated graphite felt, B-doped GF-1, B-doped GF-2 and B-doped GF-3 are presented in Fig. 6. It is seen that the coulombic efficiencies are higher than 97% for all the samples, demonstrating the good airtightness of the battery setup. Moreover, in consistent with the charge-discharge curves, the voltage efficiencies of assembled VRFBs with various electrodes decreases in sequence of B-doped GF-2 electrodes > B-doped GF-1 electrodes > B-doped GF-3 electrodes > thermally treated GF electrodes > original GF electrodes. Due to the energy efficiency is the product of voltage efficiency and coulombic efficiency, and the difference in coulombic efficiency is limited, the VRFBs with B-doped GF-2 electrodes also exhibit the highest energy efficiency among all the samples. Results show that the batteries assembled with B-doped GF-2 electrodes achieve energy efficiencies of 91.84%, 87.40% and 82.52% at the current densities of 80, 160 and 240 mA cm^{-2}, which are 11.32%, 15.63% and 19.50% higher than those with original graphite felt electrodes. More importantly, the batteries can also be operated at high current densities of 320, 400 and 480 mA cm^{-2} with the energy efficiencies 77.97%, 73.63% and 69.03%, among the highest performances in the open literature. Therefore, in consistent with our computational predictions, the boron-doped graphite felts can be the highly efficient electrodes, leading to a high rate capability for VRFBs.

In addition to the rate capability, the performances of VRFBs

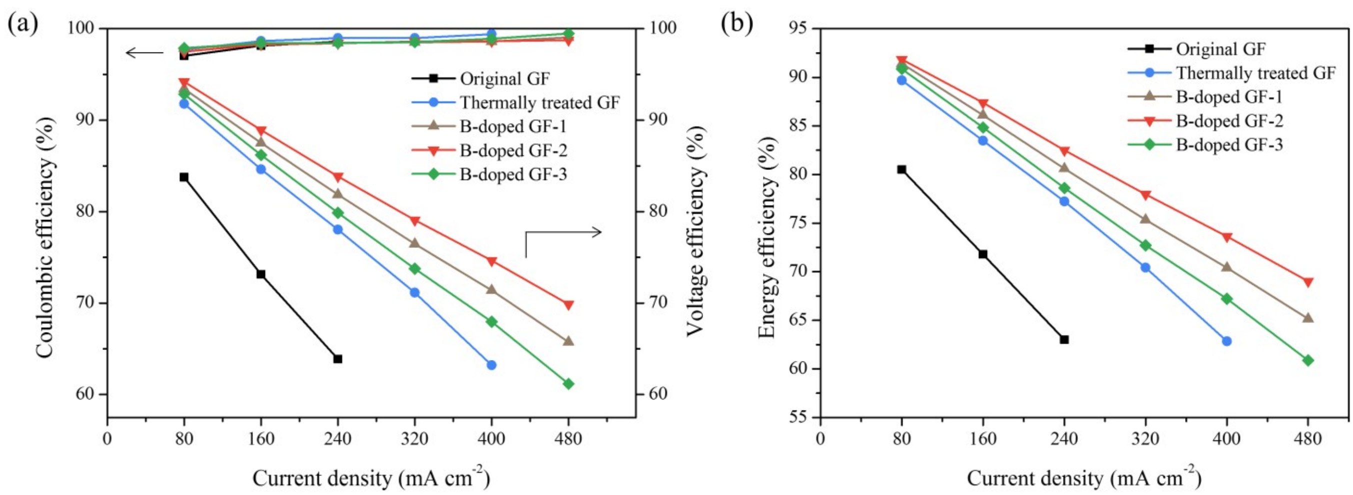

Fig. 6. The (a) coulombic efficiency and voltage efficiency, and (b) energy efficiency for VRFBs with different electrodes.

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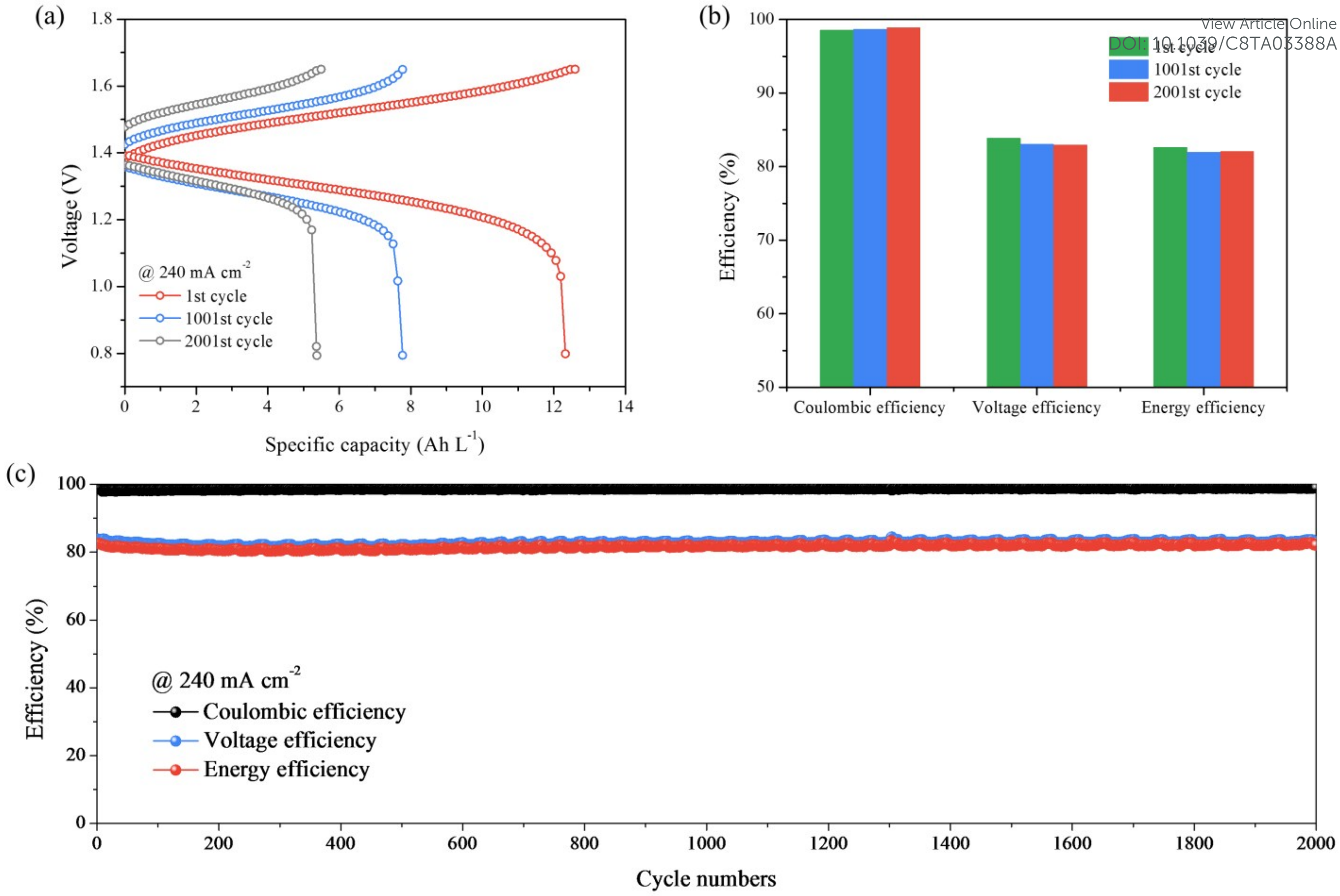

Fig. 7. The (a) charge-discharge curves and (b) efficiencies for VRFBs with boron-doped graphite felt electrodes at the 1st, 1001st and 2001st cycles during cycling. (c) The cycling performances of VRFBs with boron-doped graphite felt electrodes at the current density of 240 mA cm^{-2}.

are also depended on the cycling stability, especially considering the fact the VRFBs are developed for long-term and large-scale energy storage. In this regard, the long-term cycling tests are carried out to demonstrate the stability of the prepared electrodes, as shown in Fig. 7. Fig. 7c presents the efficiency changes of VRFBs with boron-doped graphite felt electrodes during 2000 cycles at the current density of 240 mA cm^{-2}. It is exciting to find that the battery can be stably operated for more than 2000 cycles with almost no efficiencies decay, demonstrating the excellent stability. Here, the step like fluctuation of efficiencies is caused by the temperature difference between daytime and nighttime instead of the stability of electrodes. The charge-discharge curves and efficiencies at the 1st, 1001st and 2001st cycles during cycling tests are summarized in Figs. 7a and b. Results show that the capacities decrease fast in the initial 1000 cycles, and then decrease slowly from the 1000st to 2000st cycles, which can be explained by the reduced amount of vanadium crossover due to the balanced vanadium ions in negative and positive electrolytes. The specific discharge capacity in the 1st cycle is 12.41 Ah L^{-1}, and decreases to 5.47 Ah L^{-1} in the 2001st cycle, corresponding to an ultra-low capacity decay rate of only 0.028% per cycle. In Fig. 1b, the coulombic efficiency gradually increases from the 1st to the 1001st and then increases to the 2001st cycles, due to the reduced time for vanadium ions' crossover. Oppositely, the voltage efficiency slowly decreases as the cycle numbers increases, which may be caused by the status changes of electrolyte, as well as the decay of membrane and electrodes. The energy efficiency, which is the product of coulombic and voltage efficiencies, is thus found to decrease from the 1st to the 1001st cycles, and then gradually increase from the 1001st to the 2001st cycles. In the 2001st cycle, it is surprising to find that the VRFBs with the boron-doped graphite felt

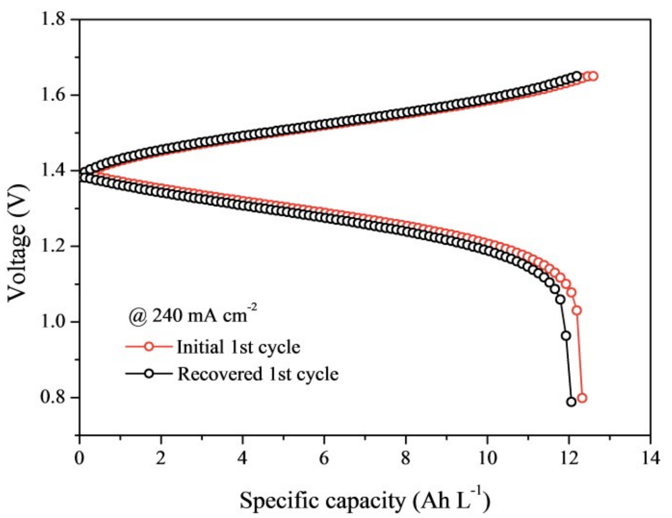

Fig. 8. The charge-discharge curves of VRFBs with boron-doped graphite felt electrodes at the initial 1st cycle and the recovered 1st cycle.

can still maintain a high energy efficiency of 82.07%, representing an ultra-low energy efficiency decay rate of 0.0002% per cycle. Then, the stability of the boron-doped graphite felt electrodes is further demonstrated by refreshing the electrolytes after 2001 cycles, the charge-discharge curves of VRFBs with boron-doped graphite felt electrodes at the initial 1st cycle and the recovered 1st cycle are shown in Fig. 8. Results clearly show that after refreshing electrolytes, the discharge-charge curves at the initial 1st cycle and the recovered 1st cycle are almost identical and the capacities are largely recovered. The slight loss of performance after refreshing electrolytes is mainly ascribed to the decay of membrane instead of electrodes, because the vanadium cations would adsorb on the sulfonate groups inside the Nafion membrane and thus the degradation of Nafion membrane is inevitable during long-term cycling tests [62]. Therefore, the boron-doped graphite felts can be regarded as the ultra-stable electrodes, ensuring the high cycling performance for VRFBs.

4. Conclusion

In this work, the highly efficient and ultra-stable boron-doped graphite felt electrodes are designed, fabricated and tested for VRFBs. The first-principles study is firstly carried out and find that the boron-doped carbon surface exhibits more a non-uniform distribution of electrons and larger energy barrier to remove reactive sites, indicating boron-doping can benefit the activity and stability of the electrodes for VRFBs. Then, the boron-doped graphite felt electrodes are fabricated and tested. CV tests demonstrate that the boron-doped graphite felt shows higher peak current density and smaller peak potential separations than the original and thermally treated graphite felts, which is indicative of the higher electrochemical performance of boron-doped graphite felt for vanadium redox reactions. In the battery tests, the VRFBs with boron-doped graphite felt electrodes achieve energy efficiencies of 87.40% and 82.52% at the current densities of 160 and 240 mA cm^{-2}, which are 15.63% and 19.50% higher than those with original graphite felt electrodes. In addition, the batteries can also be operated at high current densities of 320 and 480 mA cm^{-2} with the energy efficiencies 79.97% and 69.03%, among the highest performances in the open literature. More excitingly, the VRFBs with boron-doped graphite felt electrodes can be stably operated for more than 2000 cycles with an ultra-low capacity decay rate of 0.028% per cycle and an ultra-low energy efficiency decay rate of 0.0002% per cycle, demonstrating the excellent stability during long-term repeated charge and discharge processes. After cycling tests, the electrolytes are refreshed and then the performances of the battery are nearly recovered. All these superior results demonstrate that the boron-doped graphite felt can be a promising electrode, leading to the high activity and excellent stability for VRFBs.

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